Nguyen Hoang Anh

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Education

Sai Gon University (SGU – HCM) Expected June 2027
B.Sc. in Computer Science GPA: 3.4/4.0

Relevant Coursework

Object-Oriented Programming, Data Structures & Algorithms, Algorithm Design, Database Systems, Operating Systems, Computer Networks, Software Architecture, Discrete Math, Linear Algebra, Probability & Statistics

Skills

Languages: Python, C/C++

Frameworks & Libraries: Node.js, ReactJS, PyTorch, TensorFlow

DevOps & Tools: Docker, GitHub Actions (CI/CD), Git

Databases: MySQL, MongoDB

Tools: IntelliJ IDEA, VS Code

Projects

Neural Machine Translation System (Seq2Seq LSTM - NLP)

- Built an end-to-end neural machine translation system (English → French) using Seq2Seq LSTM.
- Implemented Encoder–Decoder architecture with custom PyTorch modules.
- Processed text data using spaCy (tokenization, vocabulary, special tokens).
- Designed data pipeline with PyTorch Dataset/DataLoader and custom padding.
- Trained model using CrossEntropyLoss, Adam optimizer, and gradient clipping.
- Implemented Greedy and Beam Search decoding strategies.
- Evaluated performance using BLEU score metrics.
- Refactored project into modular structure with configurable parameters.
- Built CLI with argparse, supporting checkpointing and early stopping.

- Visualized training progress using matplotlib.

School Bus Tracking System (Real-time Web Application)

- Developed a full-stack real-time tracking system using ReactJS, Node.js, Express, and MySQL.
- Designed SPA with role-based access (Admin, Driver, Parent) using React Router.
- Integrated Leaflet for map visualization, routing, and real-time bus movement.
- Implemented WebSocket communication (Socket.io) for live GPS tracking.
- Built secure RESTful APIs with JWT authentication, bcrypt hashing, and RBAC.
- Modeled relational database using MySQL with Sequelize ORM.
- Developed ETA-based notification system for real-time alerts.
- Created simulation tools for testing and system validation.

Sashimi POS & Sales Management System (Java Desktop Application)

- Developed a Java Swing-based POS system following 3-Tier Architecture (UI–BUS–DAO).
- Built custom UI components (modern styling, hover effects, responsive layouts).
- Applied multi-threading (SwingWorker) to handle background tasks without blocking UI.
- Designed and managed complex MySQL schema for orders, inventory, staff, and suppliers.
- Implemented core business logic: order processing, payments, promotions, and scheduling.
- Applied DTO pattern and modular architecture for scalability and maintainability.
- Integrated invoice generation and printing functionality.
- Containerized application using Docker and Docker Compose with environment configs.
- Implemented authentication and session management for secure access.